

Computer network Test

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QAns

Network layer responsibilities:

- i) logical Addressing - Assigns IP addresses to device so they can be uniquely identified
- ii) Packetizing - Encapsulates data received from the transport layer into packets.
- iii) Routing - Determines the best path for data to travel from source to destination across multiple networks
- iv) forwarding - Moves packets from the input interface to the appropriate output interface using forwarding tables.
- v) fragmentation & Reassembly - Breaks large packets into smaller fragments

Difference b/w network layer and data link layer

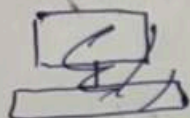
features	Network layer	Data link layer
Addressing	logical (IP Address)	Physical (MAC address)
scope	end to end (across network)	Node-to-Node (same network)
unit of data	Packet	frame
Device used	Routers	switches, Bridges
function	Routing b/w networks	Error detection within link
Protocol example	IP, ICMP	Ethernet, PPP

2) Packetizing :-

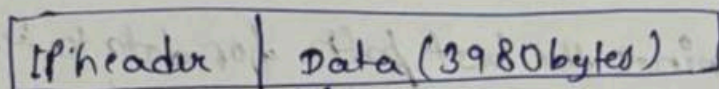
Packetizing is the process of encapsulating the payload (data received from the upper/transport layer) into network layer packets before transmission. Each packet consists of:

Header - contain source IP, destination IP, TTL, Protocol

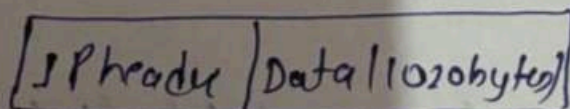
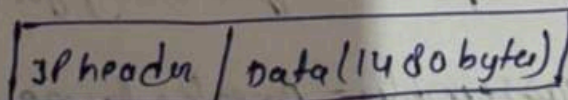
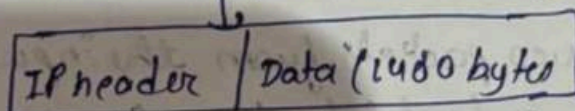
Payload - The actual data being transmitted ^{type}



Data packet (4000 bytes)



MTU = 1500 bytes



How it works

- 1) The transport layer hands data (a segment) to the network layer
- 2) The network layer wraps this data with an IP header → forming a packet
- 3) it goes to link layer, which wraps it further into a frame

Why breaking data into packets necessary

- 1) Efficient use of network - Multiple users can share the same link simultaneously
- 2) Error recovery - Only the lost packet is retransmitted, not all data
- 3) Flexible Routing - Different packets can take different paths
- 4) Size Management - Networks have MTU limits; large data must be split

3) Ans

forwarding is the router's action of receiving a packet on an input interface and sending it out through the correct output interface towards its destination. It is a fast, local per-router process

How routers use forwarding table

- 1) Packet arrives → router reads destination IP
- 2) Router checks its forwarding table using longest prefix matching
- 4) Packet is sent out through the interface.

forwarding table example

Destination	Next Hop	interface
192.168.1.0/24	10.0.0.1	eth0
172.16.0.0/16	10.0.0.2	eth1

4) Routing is the process of selecting the best path for packets to travel from source to destination across network. Routers use routing algorithm and routing tables to make these decisions.

static Routing: Routers are manually entered by the administrator. They don't change automatically.

→ Simple, secure, low overhead

Ex - small office with 2-3 routers where path never change

Dynamic Routing: Routers automatically learn and update routes using protocols like RIP, OSPF.

scalable, self-healing, adapts to failure

ex: if a link fails, OSPF recalculates a new path automatically

Feature	static	Dynamic
configuration	Manual	Automatic
fault tolerance	No	Yes

Protocols

None

RIP, OSPF

5) Datagram Network - packet delivery without fixed path

A datagram network is a packet-switched network where each packet is routed independently without pre-established fixed paths. The Internet is the best example

working

- 1) Data is split into packets, each with source & dest. IP
- 2) Each packet is forwarded independently by every router
- 3) Routers use forwarding tables to decide next stop

How delivery works without fixed path

- TTL field prevents infinite looping
- Dynamic routing tables adapt if a link fails
- Best effort delivery - no guarantee, but TCP handles reliability

Packet 1 $\rightarrow R1 \rightarrow R3 \rightarrow R5 \rightarrow \text{Destination}$

Packet 2 $\rightarrow R1 \rightarrow R2 \rightarrow R4 \rightarrow \text{Destination}$

(same message different path.)